CONCLUSION: Although the thickness is similar to the SI, structure in the OI under the subtalar joints may provide more support of the foot allowing for greater transfer of force to the ski. EMG data from TA and PL activity would indicate more efficient force transfer with OI. These muscles are critical in stabilizing the ankle and allowing for foot eversion during the edge transfer and steering phases of the turn. Time per turn was significantly lower with the OI, up to 0.02 sec/turn. With 45-60 gates per run, the OI could have a significant impact on overall performance since placings in a ski race are often separated by fractions of a second.

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200 Board #41 May 30 9:30 AM - 11:00 AM

Knee Joint-Specific Eccentric Utilization Ratio Determines Lower-Extremity Stretch-Shortening Cycle Function During Countermovement Jumps

Jordi Heeneman, John Krzyszkowski, Kristof Kipp. *Marquette University, Milwaukee, WI.* (Sponsor: Paula Papanek, FACSM)

(No relevant relationships reported)

The Eccentric Utilization Ratio (EUR) is an indicator of lower-extremity stretch-shortening cycle function in power athletes. Joint-specific EUR's have not been established, and their contributions to whole-body EUR are currently not known. Determining joint-specific EUR's and their contribution to whole-body EUR would help expand the understanding of the neuromuscular function and biomechanics of jumping exercises and provide insights for the program design process.
PURPOSE: To determine joint-specific EURs and their associations with whole-body EUR.
METHODS: Nine college track and field and soccer athletes (Height: 175 ± 15 cm, Weight: 71 ± 20 kg) participated in this study. All athletes performed three squat (SJ) and countermovement (CMJ) jumps. During all jumps, kinematic and kinetic data were obtained from 14 reflective markers (Plug-in-Gait marker set) and with two force plates, respectively. Hip, knee, and ankle joint powers of the right leg were calculated with inverse dynamics methods. In addition, jump heights were calculated from pelvis markers during each jump. EUR's (CMJ/SJ [unitless]) were calculated from the three-trial average peak joint powers of the hip, knee, and ankle joint (joint-specific EUR) and from jump heights (whole-body EUR) of the CMJ and SJ. Joint-specific EUR's were compared with a one- way ANOVA. Joint-specific EUR's were then correlated to whole-body EUR with simple linear regressions.
RESULTS: Joint-specific EUR's were 1.04±0.18, 1.15±0.25, and 1.05±0.18 for the hip, knee, and ankle joint, respectively. Joint-specific EUR's did not differ across joints. Whole-body EUR was 1.11±0.70. The Pearson correlation coefficients (r) between joint-specific EUR's of the hip, knee, and ankle joint and whole-body EUR were 0.10 ($p = 0.80$), 0.70 ($p = 0.01$), and 0.50 ($p = 0.17$), respectively.
CONCLUSIONS: The stretch-shortening cycle function of the knee joint appears to be the primary determinant of whole-body stretch-shortening cycle performance, while the ankle and hip joints do not appear to contribute much at all.

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201 Board #42 May 30 9:30 AM - 11:00 AM

Neural And Muscular Alterations Of The Plantar Flexors In Middle-aged Women

Michelle Burge, Kevin Phillips, Byungjoo Noh, Matt Gage, Tejin Yoon. *Michigan Technological University, Houghton, MI.* (Sponsor: Sandra Hunter, FACSM)

(No relevant relationships reported)

It has been recently shown that while walking, old males produced significantly less power of the plantar flexors than young males. Additionally, old males have lower absolute and relative RTD of the plantar flexors, along with smaller pennation angle, but no differences in muscle size, when compared to young men. It is of interest if similar trends are seen from young to middle age in women.
PURPOSE: Compare muscle structure and neuromuscular function of the plantar flexors in young and middle-aged women.
METHODS: Twenty two middle-aged (years: 54 ± 6) and eight young women (years: 22 ± 2) volunteered to participate in this study. This study involved 2 sessions. During the 1st session, *In vivo* muscle architecture measurements were made using a B-mode real-time ultrasound scanner. Muscle thickness and pennation angle was examined in the gastrocnemius medialis (GM) and gastrocnemius lateralis (GL) muscles. During the 2nd session, participants completed maximal voluntary isometric contractions (MVIC) of the plantar flexor muscles on a dynamometer. Single electrical stimuli were applied over tibial nerve in the popliteal fossa to elicit twitch torque during the MVIC and upon relaxation (approximately 2 s) following the MVIC to assess voluntary activation [VA = (1-superimposed twitch/resting twitch) x 100]. Maximal torque and rate of torque development were calculated and normalized by weight. Independent samples t-tests were used to examine any differences between the young and middle-aged women.
Statistical significance was set at an alpha of $p < 0.05$.

RESULTS: There were no differences in MVIC strength (128 ± 63 vs. 107 ± 30 Nm) or VA (96 ± 8.6 vs. 92 ± 10 %) between young and middle-aged women, however, rate of torque development trended towards being greater in the young women (626 ± 310 vs. 462 ± 150 Nm/s, $p = 0.06$). Additionally, no differences were noted between muscular size, however, pennation angle of the GM was significantly greater in the young women (25.8 ± 2.5 vs. 23.6 ± 2.4 degrees, $p = 0.038$).
CONCLUSION: While maximal strength of the plantar flexors are maintained in middle-aged women, there were signs of a loss in the rate of torque development, and significant changes in pennation angle of the GM muscle.

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202 Board #43 May 30 9:30 AM - 11:00 AM

Comparison Of Electromyographical Signal Analyses For Estimating Lactate Threshold

Ronald L. Snarr[1], Danilo V. Tolusso[2], Ashleigh V. Hallmark[3], Michael R. Esco, FACSM[2]. *[1]Georgia Southern University, Statesboro, GA. [2]The University of Alabama, Tuscaloosa, AL. [3]Cardiology Imaging Clinic, Birmingham, AL.* (Sponsor: Michael R. Esco, FACSM)

(No relevant relationships reported)

The relationship between the lactate and electromyographical (EMG) thresholds have previously been established via graded exercising testing during cycling. Currently, no published literature exists comparing the most appropriate and efficient filtering methods of EMG analyses to estimate the workload at which lactate threshold (LT) occurs.
PURPOSE: The purpose of this investigation was to evaluate and compare EMG transformations and time windows to predict LT.
METHODS: Participants (n=14) completed an incremental, maximal exercise test on a cycle ergometer until exhaustion. Blood lactate was measured every minute, while EMG was recorded continuously at the site of the vastus lateralis. EMG signaling was then transformed and filtered using two time-segment windows (i.e., 10 and 60 seconds), as well as three signal conversions (i.e., rootmean square, smoothing, and peak amplitude averaging).
RESULTS: Results indicated no mean differences between the EMG thresholds, for any of the filtering methods or time-segment windows, when compared to the LT criterion. Significant moderate correlations were seen when comparing the lactate and EMG time-curves ranging from 0.69 -0.79. Root mean square and Smoothing filters accurately indicated LT in 10 out of 14 participants; whereas peak amplitude averaging indicated LT for 11 out of 14 participants.
CONCLUSIONS: EMG may be a useful tool to estimate the work rate associated with LT. Averaging EMG over a minute of time and continual 10-second recordings demonstrate comparable readings and allow an easier application of EMG threshold in the field.

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203 Board #44 May 30 9:30 AM - 11:00 AM

Are Upper Body Muscle Activations Different In Various Type Of Push-up Exercise?

Melanie Poudevigne, FACSM, Moroni Demoors, Thomas Andre, Hae Chung. *Clayton State University, Morrow, GA.*

(No relevant relationships reported)

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Push-up exercises are widely used as a recommended home-based strengthening exercise for the upper body. Very little has been published on push-ups and the activation of upper limb major muscles in different hand positions using EMG. More research is needed to explain the workload done by Triceps Brachial (TB), Pectoral Major (PM) and Anterior Deltoid (AD).
PURPOSE: The aim of this study is to measure the EMG signals of TB, PM and AD while performing push-ups in two different hand positions.
METHODS: 50 African American subjects were recruited to perform push-ups on two separate occasions. The 2 different hand positions were dictated by the index or pinky alignment with the spine position. Hands were spread by shoulder length. The mean of peak EMG signals to exhaustion were analyzed. A non-parametric Kruskal-Wallis H test was applied and followed by Wilcoxon Signed-Rank test as the post hoc paired difference test. A Bonferroni correction of p=0.01 was applied which was derived from p=0.05/2.
RESULTS: Significant differences were found in all muscles with a greater activation using the index finger position in the TB muscle. Males experienced a significantly greater activation in all muscles compared to females.
CONCLUSIONS: Higher muscle activation of TB implies that the training effect of TB may be higher in the above-mentioned hand position. Future studies should address the increase in muscle strength in in-home setting especially in minorities.

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204 Board #45 May 30 9:30 AM - 11:00 AM

Changes in Bilateral Hand Force After 30 Minutes of Climbing in Elite Level Rock Climbers

Philip F. Ferrara, James Becker, John G. Seifert. *Montana State University, Bozeman, MT.*

(No relevant relationships reported)

Rock climbing is a sport that requires finger flexor strength and endurance to maintain isometric contractions during an ascent. It has been shown that climbers have higher finger flexion strength and endurance than the general population at the proximal interphalangeal joint (PIP). What is unclear, however, is if years of training and conditioning have caused these adaptations to occur bilaterally in both the dominant (D) and non-dominant (ND) arms of elite level rock climbers. It is essential for climbers to have equal, bilateral finger flexor strength and endurance capacity as rock climbing stresses both sides of the body. Weakness on one side may result in a fall or failure to complete a route.
PURPOSE: To analyze change in average force production in D and ND finger flexors of elite-level rock climbers over the course of 30 min of climbing.
METHODS: 8 elite- level (age: 29.4 ± 4.7 y, climbing experience: 11.1 ± 5.2 y, mean project grade: 9.7 ± 0.3 UIAA) rock climber's D and ND finger flexor strength was tested using a mounted force transducer. A 20-sec isometric contraction was performed with the arm fixed at 90° elbow flexion and 120° of horizontal shoulder abduction. Subjects were given two warm up contractions on each hand before the 20-sec MVIC. Subjects then climbed on a treadwall for six 5-min intervals. Total climbing time was 30 min. The treadwall rotated at 6 m/min with a 6° overhang. The isometric force assessment was repeated at the end of each interval and kept to less than 3 min to prevent recovery. Force data were analyzed by removing the first and last 10% of the contraction and averaging the remaining 16 sec. Data were expressed both in absolute (N) and as a percentage of the initial MVIC trial (relative). A paired t-test was used to compare the pre-MVIC to the MVIC at 30 min.
RESULTS: No statistical differences were found for change in absolute force between D (29.1 ± 26.1 N) and ND (41.5 ± 63.6 N) hands or in relative force between D (8.05 ± 7.5%) and ND (12.09 ± 11.7%) hand. Mean change in force relative to BW after 30 min of climbing: D (0.44 ± 0.39 N/kg BW), ND (0.62 ± 0.96 N/kg BW).
CONCLUSION: Elite-level rock climbers do not show a bilateral deficit between their D and ND hands typically seen in other sports. This could be due to years of symmetrical training and the stresses that are applied to both limbs while rock climbing.

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205 Board #46 May 30 9:30 AM - 11:00 AM

Effects Of Postactivation Potentiation On Subsequent 40-yard Sprint Performance In 16- To 23-year-old Male Athletes

Cody Yates[1], Peter J. Chomentowski[1], Mark Flury[1], Steven M. Howell[1], Anthony Deldin[2], Frank R. Wojan[1], Jamal Roper[1], Jeremy Armstrong[3].
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(No relevant relationships reported)

Postactivation potentiation (PAP) is a physiological adaptation which enables the muscles' contractile properties to optimally perform. PAP is engendered through pre- performance conditioning activities, such as parallel back squats performed prior to a vertical jump test.
PURPOSE: The purpose of this study was to determine the effects of postactivation potentiation on subsequent 40-yard sprint performance in 16- to 23-year-old male athletes, specifically, the effects of hexagonal bar deadlifts (HBD) and weighted sled sprints (WSS) as PAP-loading protocols.
METHODS: Thirty-one male subjects (age, 16.9 ± 1.4 years; height, 180.2 ± 6.2 cm; weight 83.4 ± 19.2 kg) participated in this study. Testing sessions included two different visits, a control trial and a PAP-loading protocol trial separated by ~48-hours, counterbalanced, allowing each subject to act as his own control. The HBD (n = 8) group performed four sets of HBD as the PAP-loading protocol, using body weight (BW) to calculate estimated one repetition max (1RM). The WSS (n = 23) group performed four sets of WSS for 15-yards, using

WSS loads of 25%, and 50% BW. Both PAP-loading protocols were followed by a 6-minute rest period and concluded with two laser-timed 40-yard sprint performances. Control trials for both groups consisted of identical time intervals as the PAP trial, with active movement utilized instead of the PAP-loading protocol.
RESULTS: The PAP trials had faster average 40-yard sprint times (5.35 ± 0.44 s) compared to the control trials (5.39 ± 0.39 s) for all subjects. The average difference for the PAP trials (-0.04 ± 0.10) was statistically significant (p = 0.029). However, there was statistical significance (p = 0.035) between PAP-loading groups, with WSS being the only group to improve in sprint time for the PAP trial. The WSS group improved in 40-yard sprint time for the PAP trial (5.33 ± 0.45 s) compared to the control trial (5.40 ± 0.41 s) with a PAP difference of -0.06 ± 0.10 s for 40-yard sprint time.
CONCLUSION: The use of a PAP-loading protocol enhances 40-yard sprint performance, with the use of WSS proving to generate faster sprint times compared to the HBD.
Table 2: Average Control Trial (Trial A) vs Average PAP Trial (Trial X) (mean ± SD)
Figure 3: Average control trial (trial A) vs average PAP trial (trial X)

A-42 Free Communication/Poster - **Cellular/Molecular**

Wednesday, May 30, 2018, 7:30 AM - 12:30 PM
Room: CC-Hall B

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206 Board #47 May 30 11:00 AM - 12:30 PM

Altered Arginase Activity Following Ischemic Stroke: The Role Of Metabolic Syndrome

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(No relevant relationships reported)

Animal models of human pathology are important to elucidate the underlying pathophysiological mechanisms and examine novel interventions with stroke. However, there is on-going controversy surrounding the validity of rodent models in inflammation mediated diseases. A number of successful rodent models have failed to translate the findings into clinical research. Epidemiological studies suggest the metabolic syndrome (MetS) is strongly associated with poor stroke outcomes. Our laboratory has demonstrated altered plasma arginase 1 (*ARG1*) expression in stroke patients. A*RG1 has an important role* in immune system regulation and is correlated with stroke outcome. However, how ARG activity is regulated after the stroke in a rodent model of stroke and if metabolic syndrome affects these processes are not well understood.
PURPOSE: The purpose of this study was to determine plasma ARG activity pre, and at 4hrs and 24 hrs post-stroke in lean (LZR) and obese (OZR, a model of metabolic syndrome) zucker rats. We hypothesized that OZR have altered ARG1 activity compared to LZR after the stroke.

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